

TB Geometry Reverb

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Overview

Builds a reverb from a three-dimensional shape by tracing rays inside a closed mesh from an emitter to a listener, so the model, its size, its surfaces and the two points decide the reflection pattern.

What this plug-in solves

Most reverbs offer a room size and a decay time, which describe a result rather than a place. When a sound has to belong to a specific object or an unusual space, those two numbers give no way to say what the space is. TB Geometry Reverb starts from the shape instead. It traces rays inside a closed mesh, from an emitter to a listener, and builds the reflection pattern from what they actually hit. Because the early reflections keep the timing of the real paths, the comb colouring that a shape produces is preserved rather than smoothed away. It is intended for sound design and for music where the space itself should be an audible choice.

What you can expect

- A reverb character that changes when the shape changes, not only when a decay number changes
- Early reflection timing that follows the real distances inside the model
- Audible differences between the built-in shapes and between surface materials
- A stable tail that continues while a new simulation is being prepared

How it works

The reverb is generated from a closed three-dimensional model rather than selected from a room-size menu. Rays leave the emitter, reflect from the mesh surfaces and are collected at the listener, and what they hit on the way decides the result.

Every arrival time is the real path length divided by the speed of sound, which Temperature sets. That is why Scale moves the reflections in time and why there is no separate pre-delay control.

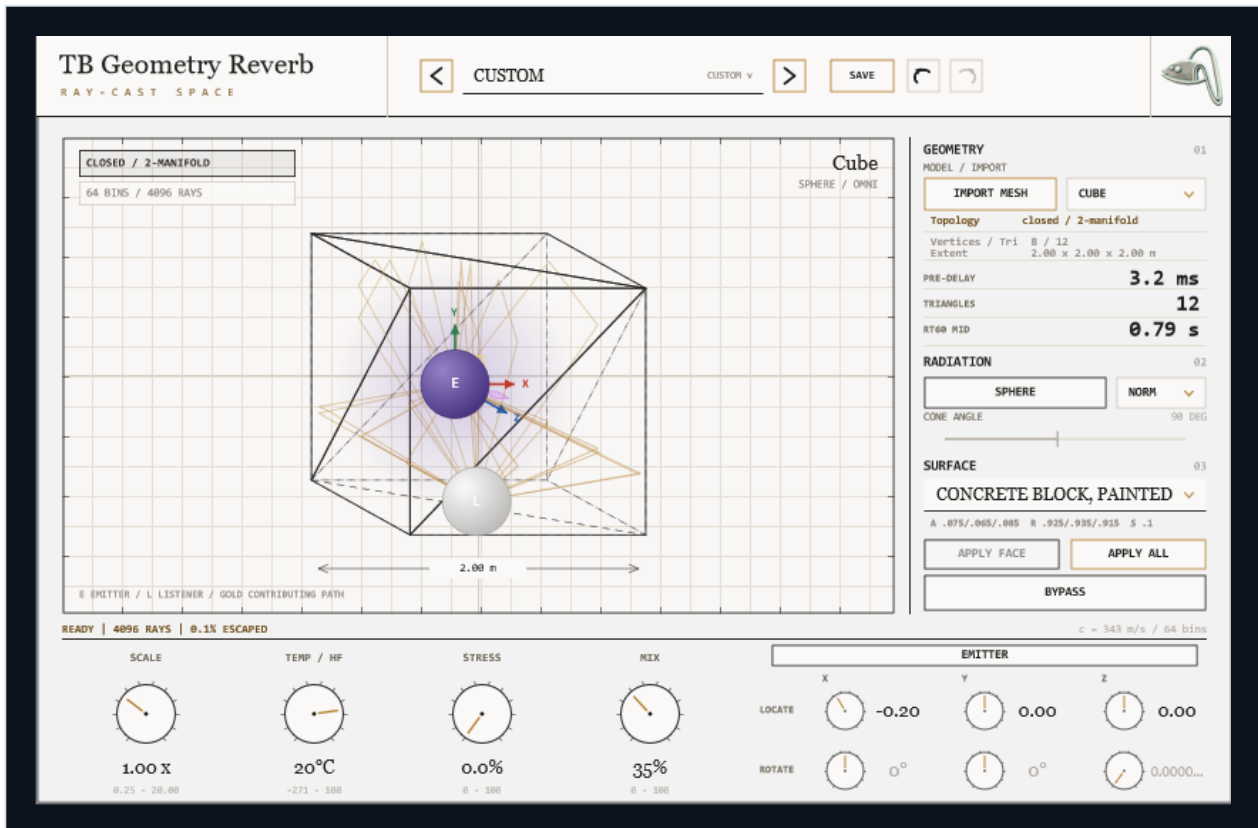
The strongest arrivals form the stereo early reflections, and the model's volume, surface area and material absorption seed an eight-line late tail. The comb colouring that discrete paths produce is kept rather than smoothed away.

All mesh parsing, validation and ray tracing happen away from the audio thread. A new simulation is crossfaded in when it is ready, and until then the previous one keeps playing.

Quick start

- 1** Insert the plug-in; the built-in Cube is simulated immediately and can be heard straight away.
- 2** Choose a shape from the model selector, or use IMPORT MESH to load your own closed OBJ or FBX.
- 3** Set Scale until the first reflection time and decay shown in the panel suit the material.
- 4** Click E and then L in the viewport and drag each point to where the source and the listener should be.
- 5** Choose Sphere for an omnidirectional source, or Cone and aim it with Emitter Azimuth and Elevation.
- 6** Assign a surface material and use APPLY ALL to hear the whole model with it.
- 7** Set Mix for the balance, then use Stress if the tail needs more presence.

Interface and key sections



This is a direct capture of the current plug-in interface. Use the visible labels to locate the areas and controls explained below.

Geometry: model and import

The GEOMETRY panel selects the shape being heard. The built-in set runs from simple solids such as Cube, Tetrahedron and Sphere through to Torus, Twisted Torus, Trefoil Knot, Hyperboloid, Superellipsoid, Capsule, Corrugated Drum, Spiked Shell and Harmonic Shell. IMPORT MESH loads your own Wavefront OBJ or FBX file. The readout beside the selector reports the topology verdict, the vertex and triangle counts and the model's extent in metres. Only a closed, two-manifold mesh can be simulated; a mesh reported as open or non-manifold is shown for diagnosis while the previous simulation keeps playing.

Scale and the metre readouts

Scale multiplies the model's dimensions. Because path times are computed from the real distances travelled, scaling the shape moves the reflections in time: a larger model spaces them further apart and a smaller one packs them together. The panel reports PRE-DELAY, which is the first reflection's arrival time in milliseconds, and RT60 MID, the estimated mid-band decay. There is no separate pre-delay or decay control, because both are consequences of the geometry.

Emitter and listener placement

The viewport shows the emitter as E and the listener as L. Clicking either point exposes its X, Y and Z arrows and its XY, XZ and YZ plane handles, so a point can be dragged along one axis or across one plane. Both points must be strictly inside the closed shell. Moving them changes which surfaces are struck first and therefore changes the early pattern, which is usually a larger difference than changing the shape's size.

Radiation: Sphere and Cone

Radiation chooses how the emitter sends sound out. Sphere radiates evenly in every direction and ignores the emitter's orientation entirely. Cone confines the rays to a beam whose width is set by Cone Angle and whose direction is set by Emitter Azimuth and Emitter Elevation. A narrow cone aimed at one wall produces a much more selective, more strongly coloured pattern than an omnidirectional source in the same shape.

Surface materials

The SURFACE panel assigns a material to the mesh. The factory library covers hard and porous masonry, brick, gypsum board, plaster, plywood, window glass, heavy velour and fibreglass panel, and the row beneath the selector shows the material's low, mid and high absorption together with its scattering value. APPLY FACE assigns the material to the selected face and APPLY ALL assigns it to the whole model. Absorptive materials shorten the decay and darken the reflections; hard materials keep them long and bright.

Quality and the simulation readout

Quality chooses the ray budget: Draft, Normal or High. Higher budgets trace more rays and give a denser, smoother early pattern at the cost of a longer rebuild. All analysis happens away from the audio thread, so raising Quality does not risk the audio stream. The status line under the viewport reports the state, the ray count and the percentage of rays that escaped, and the panel shows REBUILDING while a new simulation is being prepared.

Temperature

Temperature sets the air temperature, and the speed of sound follows it. The readout at the lower right of the viewport shows the figure in force, close to 343 metres per second at 20 degrees. Because every path time is distance divided by that speed, cooling the air lengthens the reflection times and warming it shortens them; across the full range reflections at the cold end take about twice as long to return as at the hot end. It is a timing control as much as a tonal one.

Stress

Stress drives the reverberant signal into a soft saturation instead of simply making it louder. Because the quiet detail rises relative to the peaks, the tail becomes denser and more present without the peaks running away. It is the control to reach for when a large space sounds correct but too polite. The output is compensated, so raising Stress changes the character rather than the level.

Super Grain

Super Grain decides how many of the faintest arrivals stay audible as separate events rather than falling below the floor. Raising it keeps more individual late reflections distinct, which reads as a grainier, more textured tail. At zero the control is completely inactive and the response is identical to not having the feature at all.

Mix, Output and Bypass

Mix balances the reverberant signal against the dry input, and Output trims the final level. Bypass returns the input unprocessed. When a new simulation is being prepared the previous one keeps playing and the two are crossfaded, so changing the shape, the materials or the placement does not interrupt the sound.

Presets

The header carries previous and next arrows, the current preset name, SAVE, undo and redo. Presets store the full state, including the mesh, the per-face material assignment and the material definitions themselves, so a preset recalled later cannot be changed by edits to the material library.

Controllable properties

The guide uses the names shown on the plug-in panel. Each explanation describes the audible result, a useful way to approach the control, and an important interaction to listen for.

Control	What it does and when to use it
Bypass	Returns the unprocessed input for an A/B check. Match the loudness before judging, and let the tail settle first.
Cone Angle	Sets how wide the Cone beam is. It has no effect in Sphere mode. Narrow it until one surface clearly dominates the pattern.
Emitter Azimuth	Turns the Cone beam horizontally. It has no effect in Sphere mode. Sweep it and listen for the surface the beam lands on.
Emitter Elevation	Tilts the Cone beam vertically. It has no effect in Sphere mode. Aim it at the floor or the ceiling to hear how different the same shape can be.
Listener Azimuth	Turns the listener horizontally inside the model. Turn it to change the stereo placement of the pattern without moving the listener.
Listener Elevation	Tilts the listener vertically inside the model. Small changes are enough; large ones mainly swap which surface dominates.
Listener Roll	Rotates the listener around its own facing direction. Use it to tilt the stereo image rather than to change the reflection pattern.
Mix	Balances the reverberant signal against the dry input. Set the balance here and leave Output for the final level.
Output	Trims the final output level after the reverb. Use it to match levels after the reverb, not to shape it.
Quality	Chooses the ray budget for the simulation: Draft, Normal or High. Work at Normal, then raise it once the placement is settled.
Radiation	Chooses an omnidirectional Sphere source or a directed Cone beam. Sphere ignores the emitter angles entirely; only Cone uses them.

Control	What it does and when to use it
Receiver X	Places the listener along the model's X axis. It must stay inside the closed shell. Prefer dragging L in the viewport; the arrows clamp the point to the interior for you.
Receiver Y	Places the listener along the model's Y axis. It must stay inside the closed shell. Try it near a surface and then near the centre; the difference is large.
Receiver Z	Places the listener along the model's Z axis. It must stay inside the closed shell. Separating it from the emitter along this axis widens the stereo pattern.
Rotation	Turns the model relative to the two points, changing which surfaces are struck first. Use it to find a different first-reflection surface without moving either point.
Scale	Multiplies the model's dimensions, which moves the reflections in time because path times follow the real distances. Watch the pre-delay and RT60 readouts as you set it; they tell you what the size is doing.
Source X	Places the emitter along the model's X axis. It must stay inside the closed shell. Prefer dragging E in the viewport; the arrows clamp the point to the interior for you.
Source Y	Places the emitter along the model's Y axis. It must stay inside the closed shell. Height inside the shape usually changes the first reflection more than left or right does.
Source Z	Places the emitter along the model's Z axis. It must stay inside the closed shell. Move it away from the centre so the paths to each surface become clearly unequal.
Spin X	Retained so that older sessions load. It is inert and produces no motion. Leave it alone; it exists only for session compatibility.
Spin Y	Retained so that older sessions load. It is inert and produces no motion. Leave it alone; it exists only for session compatibility.
Spin Z	Retained so that older sessions load. It is inert and produces no motion. Leave it alone; it exists only for session compatibility.

Control	What it does and when to use it
Stress	Drives the reverberant signal into a soft saturation, raising the quiet detail relative to the peaks. Reach for it when the tail is correct but too polite.
Super Grain	Sets how many of the faintest arrivals stay audible as separate events. Raise it for a grainier tail; at zero it is completely inactive.
Temperature	Sets the air temperature and with it the speed of sound, so it changes the reflection times as well as the tone. Treat it as a timing control, not only a tone control.

Practical uses

Giving a sound its own object

- Pick a shape with a strong character, such as Torus, Trefoil Knot or Corrugated Drum.
- Set Scale small so the reflections arrive early and colour the sound rather than trail behind it.
- Place the emitter and listener close to different parts of the shell so the paths are clearly unequal.
- Assign a hard material so the pattern stays audible.

Expected result: The source picks up a resonant signature that belongs to the object, not to a generic small room.

A large space that has to stay readable

- Choose a simple closed shape and raise Scale until the pre-delay readout matches the distance you want.
- Assign an absorptive material such as heavy velour or fibreglass panel to shorten the decay.
- Keep Mix modest so the dry source stays in front.
- Raise Stress only if the tail lacks presence at that Mix setting.

Expected result: The space reads as large while the source stays clear in front of it.

Directional placement inside a shape

- Switch Radiation to Cone and narrow Cone Angle.
- Aim the beam with Emitter Azimuth and Emitter Elevation and listen for the surface it strikes.
- Move the listener while the beam stays fixed to hear the pattern change.
- Raise Quality to High once the placement is settled, for a denser final pattern.

Expected result: The reverb responds to where the source is pointed, not only to where it is.

Common pitfalls

Only a closed, two-manifold mesh can be simulated. An imported model reported as open or non-manifold is displayed for diagnosis but will not replace the sound that is playing. Check the topology line beside the model selector before assuming the import failed silently; it names the fault.

The emitter and the listener must both be strictly inside the shell. Values from automation or a recalled state that fall outside it are marked invalid and block the rebuild until corrected. Drag E and L in the viewport rather than automating the raw coordinates, because the handles clamp to the interior for you.

This is a creative geometry-driven reverb, not an architectural prediction tool. The material figures are generic references, and diffraction, transmission through walls and room modes are not simulated. Choose the material for how it sounds in the model, not for what the surface is called.

The comb colouring produced by discrete paths is deliberate. If a setting sounds harsh, move the emitter or listener or change the material rather than expecting it to be smoothed away. Move one point a short distance and listen; the pattern usually changes more than any single knob does.

Spin X, Spin Y and Spin Z are retained so older sessions load, but they are inert: they produce no motion and do not trigger a new simulation. Automate Scale, Rotation or the two points instead when a session needs movement.

High Quality traces many more rays and takes longer to rebuild. The previous simulation stays audible while it works, so the delay is not a dropout. Stay on Normal while placing the points and switch to High only for the final pass.